

BALAZS HETENYI

Software Engineer | Software Developer Mentor

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PROFILE

Senior Software Developer with over five years of professional experience building secure, scalable, and user-centred applications. Skilled in Python and modern JavaScript and TypeScript frameworks, including React and Next.js, with a strong focus on performance, maintainability, and clean architecture. Experienced working within cloud-based environments and CI/CD workflows, with a commitment to code quality and thorough testing. Passionate about solving meaningful problems through thoughtful design, collaboration, and reliable engineering.

WORK EXPERIENCE

COSMIC – Wellcome Sanger Institute, Cambridge, UK (Hybrid) – Nov 2022 – Present

Senior Software Developer – Nov 2024 – Present

Software Developer – Nov 2022 – Nov 2024

- **Promoted** due to constantly exceeding expectations across technical delivery and team contribution, including mentoring junior developers.
- **Led full-stack development of a document management system** for research data ingestion, owning architecture decisions across a FastAPI/PostgreSQL backend and React frontend, coordinating feature delivery end-to-end from database schema through to UI.
- **Built and maintained a research data management platform** using FastAPI, PostgreSQL, and Elasticsearch, implementing async background task pipelines with Celery/Redis for document ingestion, PubMed literature syncing, and S3 file management enabling researchers to ingest, search, and retrieve scientific documents through a unified, automated workflow.
- **Designed an Elasticsearch sync architecture** for full-text search across scientific documents, integrating two-phase S3 upload workflows with automatic indexing triggered via Celery task queues.
- **Delivered a production-grade async Python backend** with SQLAlchemy async sessions, Alembic migrations, and Docker-compose orchestration, supporting CI/CD deployment via GitLab.
- **Developed a genomic data platform** using Next.js/React and TypeScript, featuring an AWS S3-backed secure download system with pre-signed URLs, and Elasticsearch-powered curation workflows for cancer variant data empowering clinical curators to query and manage variants from the COSMIC database through a production web application.
- **Built full-stack curation tooling** for cancer integration and actionability workflows, integrating a Django/OAuth backend with JWT token refresh, custom API client abstraction, and multi-step form validation (Yup), enabling curators to upload, query, and manage clinical genomic data.
- **Delivered production-quality frontend infrastructure:** Storybook component library, Jest/React Testing Library tests, API client, auth flows, and GitLab CI with Husky pre-commit hooks on a large-scale bioinformatics web application serving cancer genomics data.

OpenClassrooms International, UK (Part-Time Remote) – Nov 2022 – Present

Web/Software Development Mentor

- Mentored students through full-stack projects in JavaScript, React, and Node.js, guiding them from fundamentals to production-ready application architecture.
- Provided structured feedback on frontend and backend design, API integration, and deployment, with an emphasis on scalability, clean code, and professional engineering practices.

Passle Ltd., Oxford, UK (Hybrid) – May 2021 – Nov 2022

Full-Stack Software Developer

- Delivered customer-facing features across a commercial SaaS platform using Angular, TypeScript, and C#.NET, contributing to a shared codebase across multiple product teams.
- Built secure, reliable full-stack solutions in close collaboration with product and design, developing strong foundations in agile delivery and cross-functional teamwork.

EDUCATION

BSc (Honours) Computing and IT (Software Engineering) – GPA: 82% (2:1)

The Open University – Oct 2020 – Dec 2025

Foundation Degree in Software Development

OpenClassrooms International – Sept 2018 – Sept 2019

SKILLS

Frontend Development: React, TypeScript, JavaScript, React Native, Next.js

Backend Development: Python, Django, FastAPI, C#, REST API

Databases: PostgreSQL (SQL Model/ SQLAlchemy), MongoDB

Cloud & DevOps: AWS, OpenStack, Docker, Celery, GitLab, CI/CD, Supabase

Testing & Quality: Cypress, React Testing Library, Jest, Pytest, Code Reviews

State management: Zustand

Collaboration: Agile / Scrum, Mentoring, Pair Programming, Stakeholder Communication

LANGUAGES

English – Fluent

Hungarian – Native

Spanish - Beginner